THE UNIVERSITY OF HONG KONG

DEPARTMENT OF MATHEMATICS

MATH2014: Multivariable Calculus and Linear Algebra

May 10, 2019 TIME: 9:30 a.m.-12:00 noon

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Answer ALL NINE questions.

Notes:

- You should always give precise and adequate explanations to support your conclusions. Clarity of presentation of your argument counts. So **think carefully before you write.**
- You must start each question on a new page. You should write down the question number on the top right hand corner of each page.
- If your answer to a question spans over more than one page, you must indicate clearly on each page (except the last) that the answer will continue on the subsequent page.
- Whenever there is a blank page it will be assumed that all subsequent pages are also blank. If this is not true, indicate clearly on the page which you would otherwise leave blank.
- Some formulae you may need:

For $n \geq 2$:

$$\begin{aligned}
\int \cos^n \theta \, d\theta &= \frac{1}{n}\cos^{n-1}\theta \sin\theta + \frac{n-1}{n}\int \cos^{n-2}\theta \, d\theta \\
\int \sin^n \theta \, d\theta &= -\frac{1}{n}\sin^{n-1}\theta \cos\theta + \frac{n-1}{n}\int \sin^{n-2}\theta \, d\theta \\
(\arctan x)' &= \frac{1}{1+x^2}
\end{aligned}$$

1. (20 marks) In each multiple choice question below, some numbered choices are given and some are correct (wordings like is and are do not indicate the singularity /plurality of the number of correct choices). Answer the question by adding up the numbers of the correct choices. In case all choices are wrong, answer 0. Explanation is **NOT** required.

EXAMPLE: Which of the following is correct?

(1) $1 + 2 = 3$.

(2) $1 + 1 = 2$.

(4) $1/0 = 1$.

(8) $1 - 1 = 0$. Answer: 11.

(a) A system of linear equations has a unique solution. Which of the following is **NOT** possible augmented matrix of the system?

(1) $\begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 1 \end{bmatrix}$.

(2) $\begin{bmatrix} 1 & 0 & 1 & 4 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$.

(4) $\begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 3 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$.

(8) $\begin{bmatrix} 0 & 1 & 2 & 0 & 1 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$.

(b) Which of the following must be **FALSE** if matrix A is invertible?

(1) The determinant of A is zero.

(2) The rows of A are linearly independent.

(4) A is nonsingular.

(8) A row reduces to the identity matrix.

(c) For a homogeneous system of 6 linear equations with 5 unknowns, which of the following is possible?

(1) The solution set is empty.

(2) The solution set contains unique non-trivial solution.

(4) The solution set contains unique solution.

(8) The solution set contains infinitely many solutions.

(d) Which of the following matrices is **NOT** invertible?

(1) $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix}$.

(2) $\begin{bmatrix} 1 & 3 & 5 & 7 \\ 2 & 4 & 6 & 8 \\ 3 & 5 & 7 & 9 \\ 4 & 6 & 8 & 10 \end{bmatrix}$.

(4) $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 4 & 9 & 16 \\ 1 & 8 & 27 & 64 \\ 1 & 16 & 81 & 256 \end{bmatrix}$.

(8) $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{bmatrix}$.

(e) Which of the following matrices is in row echelon form?

(1) $\begin{bmatrix} 1 & -2 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

(2) $\begin{bmatrix} 1 & 0 & 2 & 3 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \end{bmatrix}$.

(4) $\begin{bmatrix} 2 & 0 & 3 & 1 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$.

(8) The 3×3 identity matrix.

(f) If A and B are both $n \times n$ matrices with rank k, which of the following must be true?

(1) $\text{rank}(A+B) = \text{rank}(B+A)$.

(2) $\text{rank}(A+B) \leq k$.

(4) $\text{rank}(A-B) \leq \text{rank}(A+B)$.

(8) $\text{rank}(A+B) = k$.

(g) Let A be a 4×3 matrix and $b \in \mathbb{R}^4$ $(b \neq 0)$, which of the following is true?

(1) $Ax = 0$ must be consistent.

(2) $Ax = b$ must be inconsistent.

(4) $Ax = 0$ may have a unique solution.

(8) $Ax = b$ may have a unique solution.

(h) A is an $n \times n$ matrix. $|A| = x$, the rank of A is y and p is one of the eigenvalues of A. If we double every entry of A and get a new matrix B, which of the following is true?

(1) $|B| = 2x$.

(2) The rank of B is $2y$.

(4) $2p$ is one of the eigenvalues of B.

(8) The reduced row echelon forms of A and B are the same.

(i) Which of the following is a linear combination of the vectors $(3,2,0)^T$ and $(0,2,3)^T$?

(1) $(3,4,3)^T$.

(2) $(3,0,-3)^T$.

(4) $(6,2,-3)^T$.

(8) $(0,0,0)^T$.

(j) Which of the following systems is consistent?

(1)

$$\begin{cases} x_1 + 2x_2 + 2x_3 = 4 \\ x_1 + 3x_2 + 3x_3 = 5 \\ 2x_1 + 6x_2 + 5x_3 = 6 \end{cases}$$

(2)

$$\begin{cases} -7x_1 - 6x_2 \quad -12x_3 = -33 \\ 5x_1 + 5x_2 \quad +7x_3 = 24 \\ x_1 \quad +4x_3 = 5 \end{cases}$$

(4)

$$\begin{cases} x_1 - x_2 \quad +2x_3 = 1 \\ 2x_1 + x_2 \quad +x_3 = 8 \\ x_1 + x_2 \quad = 5 \end{cases}$$

(8)

$$\begin{cases} 2x_1 + x_2 + 7x_3 - 7x_4 = 2 \\ -3x_1 + 4x_2 - 5x_3 - 6x_4 = 3 \\ x_1 + x_2 + 4x_3 - 5x_4 = 2 \end{cases}$$

2. (10 marks) Given that

$$A = \begin{bmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}$$

(a) (5 marks) Find all the eigenvalues and associated eigenvectors of A (Let the first non-zero entry of each eigenvector to be 1).

(b) (2 marks) Find a non-singular matrix P and a diagonal matrix D, such that $A = PDP^{-1}$.

(c) (3 marks) Find the $(3,3)$-entry of A^{2019}.

3. (10 marks) Evaluate $\int_0^1 \frac{f(x)}{\sqrt{x}} dx$, where $f(x) = \int_1^x \frac{\ln(1+t)}{t} dt$. (Keep EXACT solution.)

4. (a) (5 marks) Determine whether the function

$$f(x,y) = \begin{cases} \dfrac{x^4y^4}{(x^2+y^4)^3} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$$

is continuous at $(0,0)$ or not. Prove your result.

(b) (5 marks) Calculate $\dfrac{\partial f}{\partial x}$ where $f(x,y) = (x+2y)^{(2y-3x)}$.

5. (10 marks) Find the absolute minimum and absolute maximum value of the function $f(x,y) = \sin x + \sin y - \sin(x+y)$ on the closed region $D = \{(x,y) : x \geq 0, y \geq 0, x+y \leq 2\pi\}$.

6. (10 marks) Find the area bounded by the parabolas $y^2 = mx$, $y^2 = nx$ $(m > n > 0)$ and the lines $y = px$, $y = qx$ $(p > q > 0)$.

7. (10 marks) Evaluate the double integral $\iint_R y\,dxdy$, where R is region inside the circle $x^2 + (y-1)^2 = 1$ and outside the circle $x^2 + (y-2)^2 = 1$.

8. (10 marks) Find the volume of the region enclosed by the sphere $x^2+y^2+z^2 = R^2$ and the cylinder $x^2 + y^2 = Rx$ $(R > 0)$.

9. (10 marks) A flat circular plate has the shape of a circular disc $x^2 + y^2 \leq 1$. The temperature T at any point (x,y) on the plate, including the boundary where $x^2 + y^2 = 1$, is given by $T = x^2 + y^2 - x - y + xy$. Find the hottest and coldest points on the plate and temperature at each of these points.

******* **END OF PAPER** *******